

A basis-free Lipschitz Malliavin chain rule

Full result for arXiv:1204.2946, Remark 3.11

Source Question

Pronk and Veraar prove in [1, Proposition 3.10] that if E_0 has a Schauder basis, E_1 is UMD, $p \in (1, \infty)$, and $\phi : E_0 \rightarrow E_1$ is Lipschitz, then

$$F \in \mathbb{D}^{1,p}(E_0) \implies \phi(F) \in \mathbb{D}^{1,p}(E_1),$$

and there is a bounded linear operator

$$T_F : \gamma(H, E_0) \rightarrow L^\infty(\Omega; \gamma(H, E_1))$$

such that $D(\phi(F)) = T_F(DF)$. In Remark 3.11 they write that they do not know whether the assumption that E_0 has a basis can be avoided.

By a density and Hahn-Banach argument this yields $\iota \iota^* F = \iota \iota^* T_F(F)$. Hence we can take $\iota = T_F$. $\square$

Remark 3.11. The first part of the proof is based on the idea in [11, Proposition 5.2], where the result has been proved for Hilbert spaces E_0 and E_1 . It is surprising that their argument can be extended to a Banach space setting. We do not know if the assumption that E_0 has a basis can be avoided. In the final part of the argument in [11, Proposition 5.2] a compactness argument is used to construct an operator $T \in L^\infty(\Omega; \mathcal{L}(E_0, E_1))$ such that $D(\phi(F)) = T(DF)$. There seems to be a difficulty in this proof, and at the moment it remains unclear whether such a T exists. Note that there are subtle (measurability) differences between the latter space and $\mathcal{L}(E_0, L^\infty(\Omega; E_1))$ if E_0 and E_1 are infinite dimensional.

4. THE DIVERGENCE OPERATOR AND THE SKOROHOD INTEGRAL

Definition 4.1. Let $p \in [1, \infty)$. Let $\text{Dom}_{p,E}(\delta)$ be the set of $\zeta \in L^p(\Omega; \gamma(H, E))$ for which there exists an $F \in L^p(\Omega; E)$ such that

$$\mathbb{E} \langle \zeta, DG \rangle_{\text{Tr}} = \mathbb{E} \langle F, G \rangle_{E, E^*}, \quad G \in \mathcal{S} \otimes E^*.$$

In that case, F is uniquely determined, and we write $\delta(\zeta) = F$. The operator δ with domain

Result

If $T : G \rightarrow L^\infty(\Omega; Y)$ is bounded and $1 \leq p < \infty$, write $T^{[p]} : L^p(\Omega; G) \rightarrow L^p(\Omega; Y)$ for the canonical amplification defined on simple random variables by

$$T^{[p]} \left(\sum_{j=1}^m \mathbf{1}_{A_j} R_j \right) = \sum_{j=1}^m \mathbf{1}_{A_j} T R_j$$

and then extended by continuity. This is the meaning of $T_F(DF)$ used below, and is the one employed in the proof of [1, Proposition 3.10].

Theorem 1. Let H be a separable Hilbert space, let $p \in (1, \infty)$, let E_0 be an arbitrary Banach space, and let E_1 be a UMD Banach space. If $\phi : E_0 \rightarrow E_1$ is Lipschitz with constant L , then for every $F \in \mathbb{D}^{1,p}(E_0)$ one has

$$\phi(F) \in \mathbb{D}^{1,p}(E_1).$$

Moreover

$$\|D\phi(F)(\omega)\|_{\gamma(H, E_1)} \leq L \|DF(\omega)\|_{\gamma(H, E_0)} \quad \text{for a.e. } \omega \in \Omega,$$

and there exists a bounded linear operator

$$T_F : \gamma(H, E_0) \rightarrow L^\infty(\Omega; \gamma(H, E_1))$$

with $\|T_F\| \leq L$ such that

$$D(\phi(F)) = T_F^{[p]}(DF).$$

Corollary 1. *The Schauder-basis assumption in [1, Proposition 3.10] can be removed. No separability or approximation property of E_0 is needed.*

Proof Intuition

The source proof uses a basis to approximate the identity on E_0 by uniformly bounded finite-dimensional projections. This is stronger than the chain rule needs. A Malliavin-Sobolev random variable is already obtained as a limit of smooth random variables with finite-dimensional ranges. We can apply the finite-dimensional Lipschitz chain rule to those approximants and pass to the limit.

The remaining operator T_F is only a factorization through the particular random derivative DF . Once the pointwise estimate $\|D\phi(F)\| \leq L\|DF\|$ is known, the strong measurability of DF puts its essential range in a separable subspace of $\gamma(H, E_0)$. On that separable subspace we build a measurable scalar norming map. The standard Goodner–Kelley–Nachbin injectivity theorem for scalar L^∞ spaces extends this scalar map to all of $\gamma(H, E_0)$. Multiplication by the normalized derivative $D\phi(F)/\|DF\|$ gives the desired T_F .

A finite-dimensional lemma

Lemma 1. *Let V be a finite-dimensional Banach space, let E_1 be UMD, and let $\psi : V \rightarrow E_1$ be Lipschitz with constant L . If $G \in \mathbb{D}^{1,p}(V)$, then $\psi(G) \in \mathbb{D}^{1,p}(E_1)$ and*

$$\|D\psi(G)(\omega)\|_{\gamma(H, E_1)} \leq L\|DG(\omega)\|_{\gamma(H, V)} \quad a.s.$$

Proof. Identify V linearly with $\mathbb{R}^m$ and convolve ψ with a standard smooth mollifier on V . This gives C^∞ maps $\psi_\varepsilon : V \rightarrow E_1$ such that

$$\|\psi_\varepsilon(x) - \psi(x)\| \rightarrow 0 \quad (x \in V), \quad \|\psi'_\varepsilon(x)\|_{\mathcal{L}(V, E_1)} \leq L.$$

The smooth chain rule [1, Proposition 3.9] gives

$$\psi_\varepsilon(G) \in \mathbb{D}^{1,p}(E_1), \quad D\psi_\varepsilon(G) = \psi'_\varepsilon(G)DG,$$

and the ideal property of γ -radonifying operators gives the pointwise bound

$$\|D\psi_\varepsilon(G)(\omega)\|_{\gamma(H, E_1)} \leq L\|DG(\omega)\|_{\gamma(H, V)}.$$

Also $\psi_\varepsilon(G) \rightarrow \psi(G)$ in $L^p(\Omega; E_1)$. Since E_1 is UMD, $\gamma(H, E_1)$ is reflexive, so the compactness lemma [1, Lemma 3.8] implies $\psi(G) \in \mathbb{D}^{1,p}(E_1)$, after passing to a weakly convergent derivative subsequence.

Mazur's lemma gives convex combinations of the derivatives $D\psi_{\varepsilon_j}(G)$ which converge strongly in $L^p(\Omega; \gamma(H, E_1))$ to $D\psi(G)$. The same pointwise bound is preserved by convex combinations. Passing to an almost surely convergent subsequence yields the asserted pointwise estimate. $\square$

The membership and pointwise bound

Choose smooth random variables $F_n \in \mathcal{S} \otimes E_0$ such that

$$F_n \rightarrow F \quad \text{in } \mathbb{D}^{1,p}(E_0).$$

Each F_n takes values in a finite-dimensional subspace $V_n \subseteq E_0$. Applying Lemma 1 to $\phi|_{V_n} : V_n \rightarrow E_1$ gives

$$\phi(F_n) \in \mathbb{D}^{1,p}(E_1), \quad \|D\phi(F_n)(\omega)\|_{\gamma(H,E_1)} \leq L \|DF_n(\omega)\|_{\gamma(H,E_0)}.$$

Since ϕ is Lipschitz,

$$\phi(F_n) \rightarrow \phi(F) \quad \text{in } L^p(\Omega; E_1).$$

The derivatives $D\phi(F_n)$ are bounded in

$$L^p(\Omega; \gamma(H, E_1)),$$

so the compactness lemma again gives $\phi(F) \in \mathbb{D}^{1,p}(E_1)$ and, after passing to a subsequence,

$$D\phi(F_n) \rightharpoonup D\phi(F) \quad \text{in } L^p(\Omega; \gamma(H, E_1)).$$

It remains to keep the pointwise estimate. By Mazur's lemma there are convex combinations

$$Z_k = \sum_{n \geq k} a_{k,n} D\phi(F_n)$$

which converge strongly in $L^p(\Omega; \gamma(H, E_1))$ to $D\phi(F)$. Set

$$M_k = \sum_{n \geq k} a_{k,n} \|DF_n\|_{\gamma(H,E_0)}.$$

Since $DF_n \rightarrow DF$ in $L^p(\Omega; \gamma(H, E_0))$, we have

$$M_k \rightarrow \|DF\|_{\gamma(H,E_0)} \quad \text{in } L^p(\Omega).$$

For every k ,

$$\|Z_k(\omega)\|_{\gamma(H,E_1)} \leq LM_k(\omega) \quad \text{a.s.}$$

Taking an almost surely convergent subsequence of Z_k and M_k gives

$$\|D\phi(F)(\omega)\|_{\gamma(H,E_1)} \leq L \|DF(\omega)\|_{\gamma(H,E_0)}$$

for almost every ω .

A factorization lemma

Lemma 2. *Let G and Y be Banach spaces, let $1 \leq p < \infty$, and let $X \in L^p(\Omega; G)$, $Z \in L^p(\Omega; Y)$ be strongly measurable. Assume*

$$\|Z(\omega)\|_Y \leq L \|X(\omega)\|_G \quad \text{for a.e. } \omega.$$

Then there is a bounded linear operator

$$T : G \rightarrow L^\infty(\Omega; Y)$$

with $\|T\| \leq L$ such that $T^{[p]}X = Z$.

Proof. By strong measurability there is a closed separable subspace $S \subseteq G$ such that $X(\omega) \in S$ for almost every ω . On S choose a Borel norming selection

$$J : S \rightarrow B_{S^*}, \quad Jx(x) = \|x\|_S.$$

This is the standard Kuratowski–Ryll–Nardzewski selection applied to $x \mapsto \{x^* \in B_{S^*} : x^*(x) = \|x\|\}$; separability makes the weak-star compact ball B_{S^*} metrizable on the relevant coordinates.

Define $A : S \rightarrow L^\infty(\Omega)$ by

$$(AR)(\omega) = J(X(\omega))(R), \quad R \in S.$$

Then A is linear and $\|A\| \leq 1$. The scalar space $L^\infty(\Omega)$ is 1-injective by the Goodner–Kelley–Nachbin extension theorem, so A has an extension

$$\tilde{A} : G \rightarrow L^\infty(\Omega)$$

with $\|\tilde{A}\| \leq 1$.

Put

$$W(\omega) = \begin{cases} Z(\omega)/\|X(\omega)\|_G, & X(\omega) \neq 0, \\ 0, & X(\omega) = 0. \end{cases}$$

Then $W \in L^\infty(\Omega; Y)$ and $\|W\|_\infty \leq L$. Multiplication by W defines a bounded map

$$M_W : L^\infty(\Omega) \rightarrow L^\infty(\Omega; Y), \quad b \mapsto bW,$$

with $\|M_W\| \leq L$. Set

$$T = M_W \tilde{A}.$$

Then $T : G \rightarrow L^\infty(\Omega; Y)$ is linear and $\|T\| \leq L$.

Let X_n be simple S -valued random variables with $X_n \rightarrow X$ in $L^p(\Omega; S)$ and almost surely. Since $\tilde{A}$ agrees with A on S , the definition of $T^{[p]}$ gives

$$(T^{[p]}X_n)(\omega) = J(X(\omega))(X_n(\omega))W(\omega) \quad \text{a.s.}$$

Therefore

$$\|T^{[p]}X_n - Z\|_{L^p(\Omega; Y)} \leq L\|X_n - X\|_{L^p(\Omega; G)} \rightarrow 0,$$

because $J(X(\omega))(X(\omega))W(\omega) = Z(\omega)$, with both sides zero on $\{X = 0\}$. Since $T^{[p]}$ is bounded on $L^p(\Omega; G)$, this proves $T^{[p]}X = Z$. $\square$

Proof of Theorem 1

Apply Lemma 2 with

$$G = \gamma(H, E_0), \quad Y = \gamma(H, E_1), \quad X = DF, \quad Z = D\phi(F).$$

The pointwise estimate proved above gives the hypothesis of the lemma. The resulting operator T is the required T_F .

Novelty Check and Scope

The argument answers exactly the operator-valued statement of [1, Proposition 3.10]: T_F lies in $\mathcal{L}(\gamma(H, E_0), L^\infty(\Omega; \gamma(H, E_1)))$. It deliberately does not claim the stronger and different object $T \in L^\infty(\Omega; \mathcal{L}(\gamma(H, E_0), \gamma(H, E_1)))$. Remark 3.11 already notes that such random-operator-field formulations have additional measurability subtleties.

The run indexes were searched for arXiv:1204.2946 and the relevant Malliavin, UMD, Lipschitz-chain-rule, basis, and bounded-approximation-property terms. Targeted web searches on 2026-06-23 found the source paper but no later exact answer to Remark 3.11 or to the basis-free strengthening proved here.

References

- [1] M. Pronk and M. Veraar, *Tools for Malliavin calculus in UMD Banach spaces*, Potential Analysis 40 (2014), 307–344; arXiv:1204.2946.
- [2] J. L. Kelley, *Banach spaces with the extension property*, Transactions of the American Mathematical Society 72 (1952), 323–326.